Progress Report Sprint 7

- Course Name: CS461 Senior Software Engineering Project I
- Date: 01/30/26
- Team Name: TaxonBodyMassML
- Team ID: CS.032
- Student Names and Emails:

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Summary - Outcomes, blockers, and insights

- **UI: Gained feedback from project partner on last week's graphic and tutorial idea; Updated the graphic based on project partner feedback; Implemented tutorial / tool tip feature on the website in order to aid usability**

- **The first version of the decision tree regression is complete, however it still needs some work.**

- **Communication was smoother this sprint and meeting with the project partner was a great boost in ideas and in morale**

Blockers:

- **Faced a barrier regarding the tutorial feature after talking with project partner, who suggested traditional tool tips may be annoying to the average user; overcame this blocker by talking with and getting advice from a specialist in web design**
- **We had an issue with the API used to gather taxonomy information. For many species, it was unable to find the one or both of the Class/Order. We plan to resolve this by cross referencing another taxonomy database.**

Work Log

1. TaxonBodyMassML Graphic

Status: In-progress

Owner: Hailey

Description: Last sprint, I mocked a graphic to use on our website to represent the TaxonBodyMassML and to promote visual interest in our website. This sprint, I received project partner feedback on the graphic and updated it accordingly.

Source Link: 2026-01-13-CS.032-Sprint6-ProgressReport.docx

Supporting Artifact: Item.01

Notes: Now, since I've gotten approval from our project partner, I can proceed in actually drawing up the mocked graphic.

2. **Tutorial Skeleton**

Status: Complete

Owner: Hailey

Description: To aid in the usability of our website, I wanted to introduce a tool tip / tutorial feature. After seeking the advice of a recent OSU graduate who specialized in web development and design, I settled on a tutorial button that the user could use to "opt in" to view the tool tips rather than making them mandatory. I coded in all the elements and movements required to implement the feature.

Source Link: 2026-01-13-CS.032-Sprint6-ProgressReport.docx

Supporting Artifacts: Item.02

Notes: The pop-ups still need styling, but the skeleton is complete.

3. **Decision Tree Regression**

Status: In-progress

Owner: Grant

Description: The first version of the decision tree regression mass prediction model is complete. It uses the decision tree regression from the xgboost library. For most of the test set species above 1 g, the prediction was pretty accurate. However, there seems to be a problem with correctly guessing the mass of species that are extremely small.

Source Link: 2026-01-13-CS.032-Sprint6-ProgressReport.docx

Supporting Artifacts: Item.04

Notes: For information about the xgboost library, check out the manpages here: https://xgboost.readthedocs.io/en/stable/

Risk and Quality Tracking

Open Bug Count: 5

Top Bugs	Severity	Owner(s)	Status	Link	Actions
Dataset Too Large For Decision Tree	High	Grant	In-progress	https://github.com/marknovak/TaxonBodyMassML/tree/random-forest/predictive_models/decision_tree_test.py	Since the model worked with about 30% of the dataset, as soon as the rest of the data is fixed, Grant will try running the model on a larger portion of the data.
Interconnected JS Movements	Low	Hailey	In-progress	https://github.com/marknovak/TaxonBodyMas	Hailey will comb through the JS code and remove any movements that solely rely on classes of other

				sML/blob/main/web_dev/index.js	elements by the end of Sprint 8
Non-responsive CSS	Low	Hailey	In-progress	https://github.com/marknovak/TaxonBodyMassML/blob/main/web_dev/style.css	Hailey will add in CSS that better accommodates smaller screen sizes (media queries) by the end of this term (not a top priority)
Mismatched Element IDs (Javascript)	Low	Hailey	In-progress	https://github.com/marknovak/TaxonBodyMassML/blob/web-dev/web_dev/index.js	Hailey will go through the tutorial segment and realign the ID names to their associated elements
Missing Class/Order From Many Dataset Rows	High	Grant	In-progress	https://github.com/marknovak/TaxonBodyMassML/tree/data-combination/data-combination	Grant will reference a different taxonomy API to fill in missing data values.

Next Sprint Plan

- Continue work on CSS from sprint 6 (including finishing the tutorial styling and the graphic) to help with UI accessibility and past CSS sprint goals.
  Priority: Medium; Link: 2026-01-13-CS.032-Sprint6-ProgressReport.docx
- Clean up the 'FAQ' section of the help page and add more content to the page to aid in usability and to accommodate users with differing levels of technical literacy.
  Priority: Medium; Link: 2025-10-23-CS.032-Requirements (REQ-004)
- After using another taxonomy dataset to fill in any missing taxonomy information, we will run the existing model on a larger dataset. Then, we will also focus on tweaking the model to achieve higher accuracy. This will help us achieve the accuracy requirement:
  Priority: High; Link: 2025-10-23-CS.032-Requirements (REQ-008)

Team Reflection

This week, we fought to meet our goals of better communication from last sprint. We stayed in contact with our project partner on Slack even after our meeting, and stayed consistent in our communication with each other. Getting feedback from Mark really made a difference this sprint; the UI got some touch-ups and we got feedback on the API we began implementing last sprint.

There was a little bit of a barrier facing the implementation of a tutorial on our website because Mark pointed out that required pop-ups can be very annoying for users. As a result, a third-party web design specialist was consulted and we moved past the problem. The things we want to improve on are our work toward our two algorithms; there hasn't been much of a chance to get good progress done because of the work we had to do on the data, but that can change now. In order to keep ourselves accountable, we're making a goal to have some code pushed to the GitHub repo by the end of the next sprint. Overall, we are pleased with the progress made this sprint and feel like we really have the groundwork needed to make substantial updates in the next couple of weeks.

Artifacts Appendix

Item.01

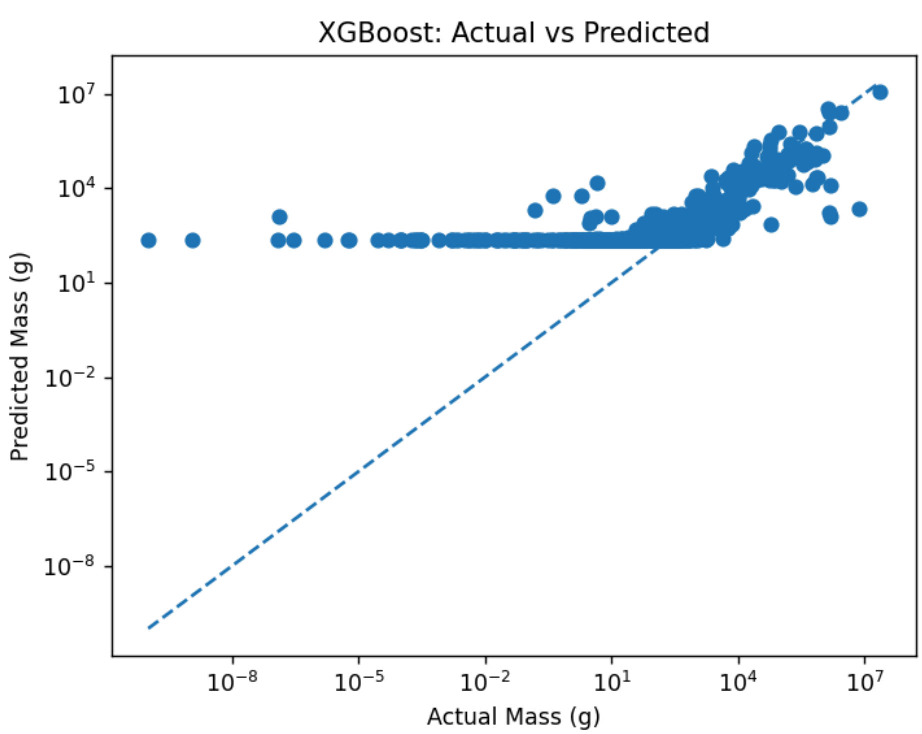
This is the new mocked-up graphic after feedback from the project partner, made on Canva. The changes focus around the incorporation of the aquatic species seen under the title text. This was done to meet the project partner's request for broader species representation.

Item.02

Screencast from 2026-02-01 19-39-52.webm

This is a screen recording of the current implementation of the UI's tutorial feature. Since the user interface is mostly complete, the contents of the tutorial will be unlikely to change much. However, slight variations may occur.

Item.03



This is a log-scale plot of the mass predictions for species in the test dataset. Evidently, work needs to be done to improve testing accuracy on species with near-zero mass. One of the immediate steps to achieve this will be solving our missing Class/Order data, so we can produce a much larger model.